Chapter test

17 Coordination in humans

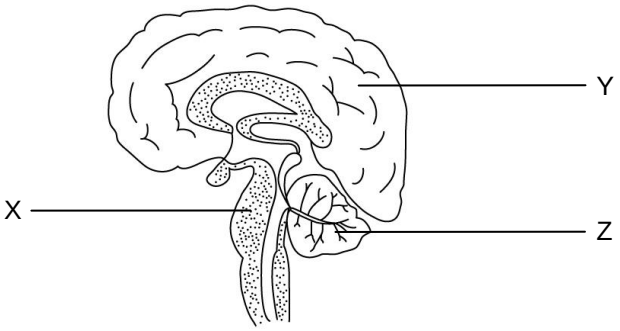
Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

Directions: Questions 1 and 2 refer to diagram below, which shows a section of a human brain.



1. Which parts of the brain carry out the functions below?

	Control of heart beat	*Muscular coordination*
A.	X	Y
B.	X	Z
C.	Z	X
D.	Z	Y

☐

★ 2. Which of the following statements about structure Y and Z is / are correct?

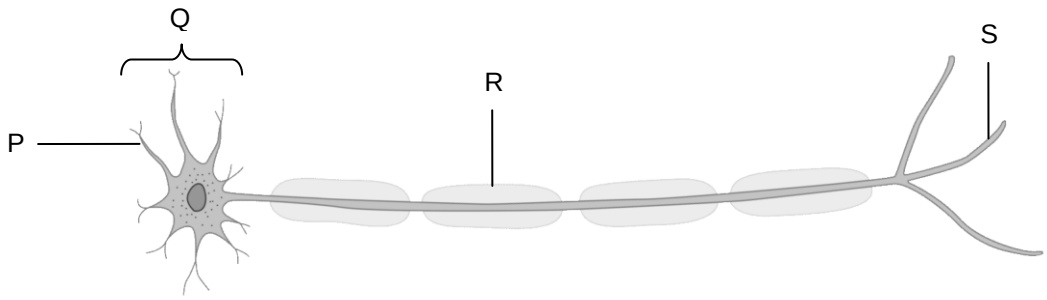
(1) Both have grey matter on the outside, and white matter inside.
(2) Both are nourished by the cerebrospinal fluid.
(3) Both receive nerve impulses from the ears.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

☐

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★ 3. The diagram below shows the structure of a motor neurone.



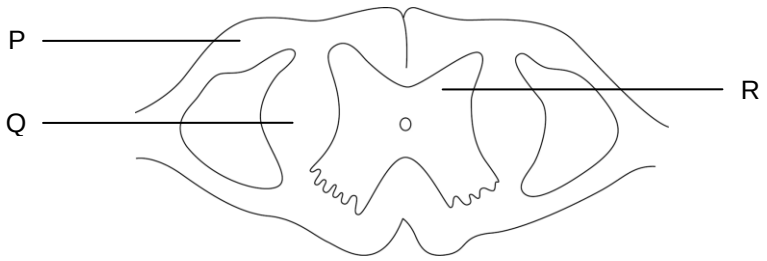
Which of the following statements about the motor neurone is ***incorrect***?

A. Nerve impulses are transmitted from P to S.
B. Synapses can be found at the end of both P and S.
C. Q is found outside the central nervous system.
D. R speeds up the transmission of nerve impulses. ☐

4. Which of the following does ***not*** belong to the central nervous system?

A. optic nerve
B. cerebellum
C. medulla oblongata
D. spinal cord ☐

5. The diagram below shows a transverse section of the spinal cord.

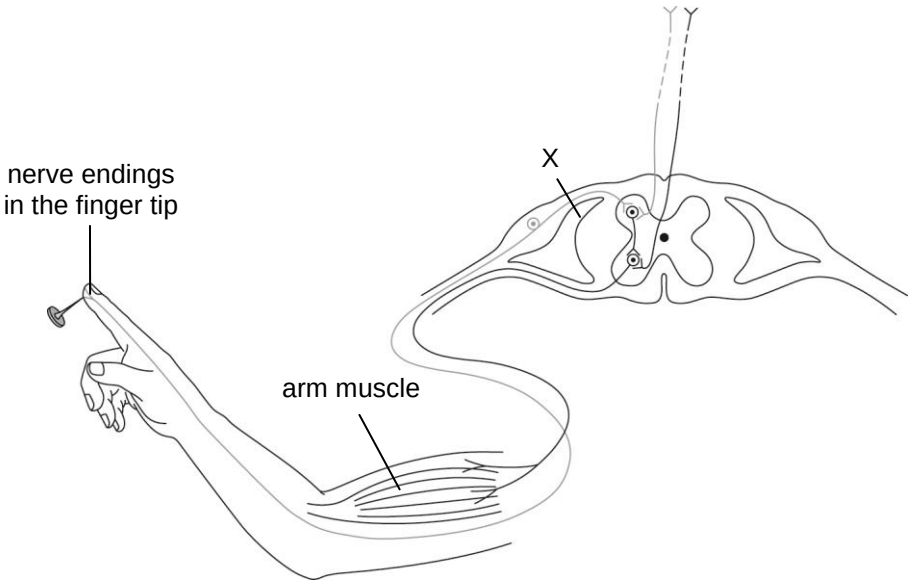


Synapses can be found in

A. P only.
B. R only.
C. P and R only
D. P, Q and R. ☐

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Directions: Questions 6 and 7 refer to diagram below, which shows the nervous pathway involved in the withdrawal reflex when a person pricks a finger on a pin.



6. How many neurones are involved in the reflex arc of the withdrawal reflex?

A. 2
B. 3
C. 4
D. 5 ☐

★ 7. In an accident, the dorsal root of the person was cut at X. What would happen if the person was pricked when blindfolded?

	Feel the pain	***Withdraw his arm immediately***
A.	yes	yes
B.	yes	no
C.	no	yes
D.	no	no ☐

8. Which of the following comparisons between voluntary actions and reflex actions is ***incorrect***?

	Voluntary actions	***Reflex actions***
A.	not inborn	inborn
B.	involve the cerebrum	do not involve the cerebrum
C.	in response to internal stimuli	in response to external stimuli
D.	responses may vary	responses are always the same ☐

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★ 9. Which of the following about the endocrine glands is correct?

	Endocrine gland	*Hormone produced*	*Target organ*	*Effect*
A.	adrenal gland	adrenaline	heart	decreases heart rate
B.	pituitary gland	luteinizing hormone	uterus	ovulation
C.	liver	glucagon	body cells	increases blood glucose
D.	pancreas	insulin	liver	decreases blood glucose

10. Which of the following statements about hormones is / are correct?

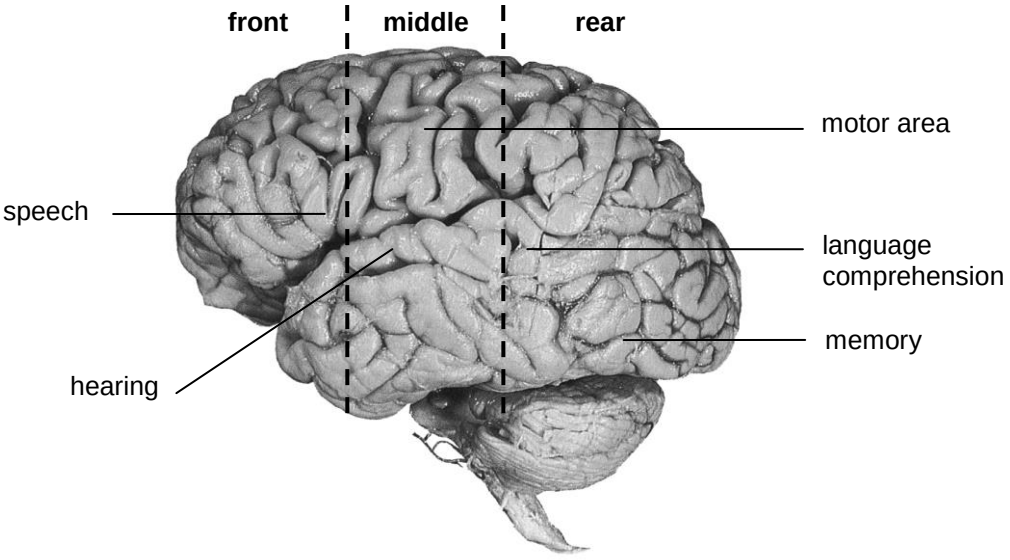
(1) All hormones are proteins.
(2) They are transported in the blood.
(3) Each hormone only acts on one target organ.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

Question	1	2	3	4	5	6	7	8	9	10
Related section	17.3	17.3	17.2	17.1	17.3	17.4	17.4	17.4	17.5	17.5

B. Problem solving (20 marks)

★ 11. The diagram below shows the lateral view of a human brain and the location of some functional areas of the cerebral cortex.



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(a) The cerebral cortex is highly folded. What is the significance of this feature? (2 marks)

(b) Cerebrospinal fluid surrounds the brain and fills the cavities inside the brain. State ***two*** functions of the cerebrospinal fluid. (2 marks)

(c) Hydrocephalus is a condition in which excess cerebrospinal fluid builds up within the cavities of the brain, putting pressure on the brain and causing damage to neurones. The table below shows how hydrocephalus affects the account of neurones in different parts of the cerebrum.

	Amount of neurones as a percentage of cerebrum volume	
Part of the cerebrum	**People with hydrocephalus**	**Unaffected people**
Front	90.7	88.8
Middle	85.3	90.4
Rear	84.0	90.7

People with hydrocephalus may show the following features:

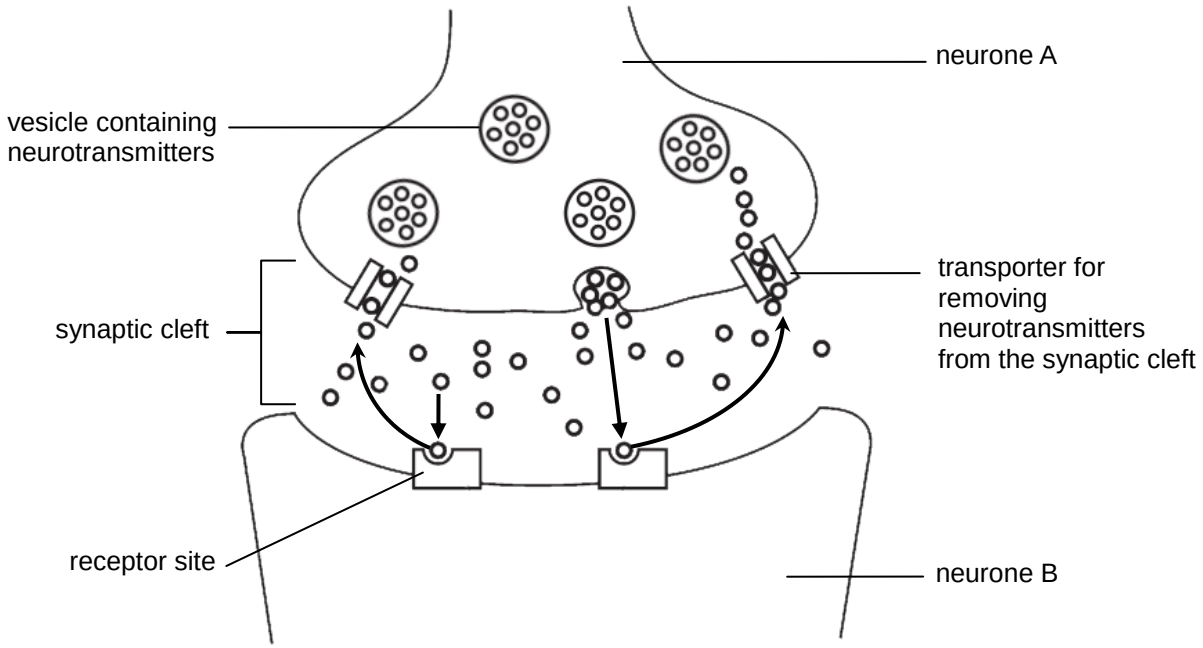
- loss of fine motor skills
- poor memory
- poor understanding of written and spoken language
- normal hearing
- normal speech production

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Using the information provided, explain the features observed in people with hydrocephalus. (6 marks)

(Total: 10 marks)

★★ 12. The diagram below shows the junction between two adjacent neurones, A and B.



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(a) Describe how a nerve impulse is transmitted across the synapse. (6 marks)

(b) This synapse only transmits information from neurone A to neurone B and not from B to A. Explain why. (2 marks)

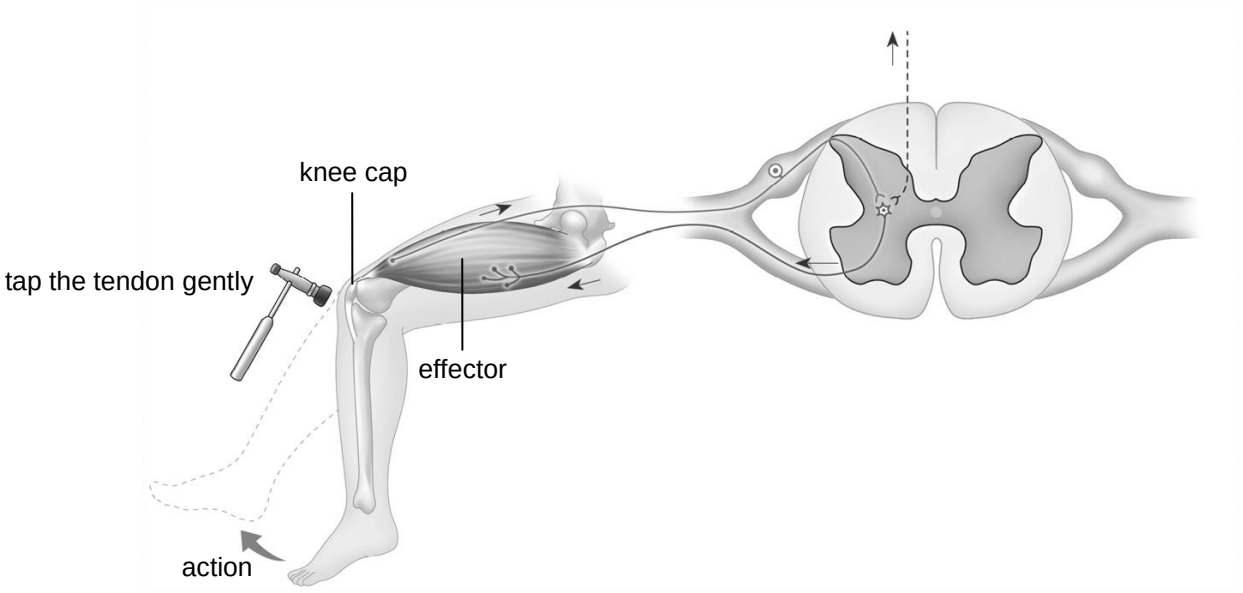
(c) For the proper functioning of the synapse, neurotransmitters have to be removed from receptor sites and from the synaptic cleft. Explain why. (2 marks)

(Total: 10 marks)

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C. Exam challenge (10 marks)

★★ 13. The diagram below shows the reflex arc of the knee jerk reflex.



(a) In the knee-jerk reflex, a light tapping (the stimulus) is detected by a stretch receptor which passes nerve impulses to the sensory neurone. The effector is the thigh muscle.

(i) Draw and label the structure of a sensory neruone. (3 marks)

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(ii) What happens to the effector in producing the response of this reflex? (1 mark)

(b) Two men (X and Y) were injured in an accident. The doctor found out that their brains were functioning normally. He conducted further tests to check if there was any damage to parts of the nervous pathway. The results of the tests are shown below.

	Test(s) conducted with the patient blindfolded	**Observations**
X	The tendon just below the knee cap was tapped.	X felt the tapping but he did not kick his leg.
Y	The tendon just below the knee cap was tapped.	Y kicked his leg forwards but he was unaware of the tapping and his kicking action.

(i) Why were the patients blindfolded during the tests? (1 mark)

(ii) Which part of the nervous pathway was most likely damaged in X and Y respectively? Explain the observations in each case. (5 marks)

(Total: 10 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	B	2.	D	3.	C	4.	A	5.	B
6.	B	7.	D	8.	C	9.	D	10.	A

(1 mark each)

B. Problem solving (20 marks)

11. (a) This increases the surface area (1)
so that a greater number of neurones can be packed in this region. (1)

(b) Any ***two*** of the following: (2)
- It supplies nutrients / oxygen to the brain cells. (1)
- It removes wastes from the brain cells. (1)
- It maintains the shape of the brain. (1)
- It serves as a shock absorber. (1)

(c) There is the largest loss of neurones / damage in the rear part of the cerebrum, (1)
leading to poor memory and poor understanding of written and spoken language. (1)
There is some (less) loss of neurones / damage in the middle part of the cerebrum, (1)
affecting motor control but not hearing. (1)
There is no damage in the front part of the cerebrum, (1)
so speech is not affected. (1)

(Total: 7 marks)

12. (a) Upon stimulation by a nerve impulse, (1)
vesicles containing neurotransmitters fuse with the membrane of neurone A (the presynaptic neurone). (1)
Neurotransmitters are released into the synaptic cleft. (1)
Neurotransmitters diffuse across the synaptic cleft down a concentration gradient. (1)
Neurotransmitters bind to receptor sites on the membrane of neurone B (the postsynaptic neurone). (1)
This stimulates the generation of a nerve impulse in neurone B. (1)

(b) Only the axon of neurone A has synaptic vesicles that can release neurotransmitters into the synaptic cleft. (1)
Only the dendron of B has receptor sites to bind with the neurotransmitters. (1)

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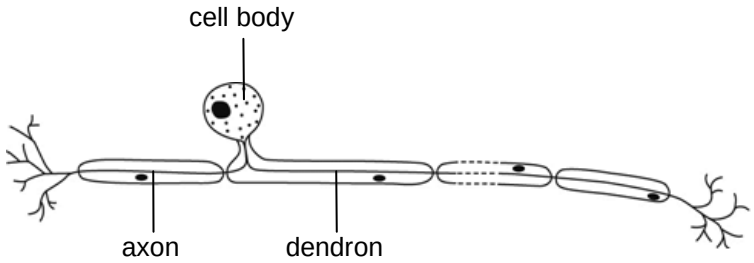
(c) Any ***two*** of the following: (2)

- To maintain a concentration gradient of the neurotransmitters for diffusion across the synaptic cleft. (1)
- To free the receptor sites to bind with more neurotransmitters. (1)
- To recycle the neurotransmitters. (1)

(Total: 10 marks)

C. Exam challenge (10 marks)

13. (a) (i) Round cell body correctly labelled (1)
Short axon correctly labelled (1)
Long dendron correctly labelled (1)



(ii) It contracts. (1)

(b) (i) To avoid the reflex action from being affected by voluntary actions. (1)

(ii) X: The motor neurone was damaged. (1)
No nerve impulses could be transmitted to the effector (thigh muscle) to produce the response. (1)

Y: The interneurone to the brain was damaged. (1)
No nerve impulses could be carried to the brain to produce the sensation. (1)
Since Y could kick his leg, the reflex arc was not damaged. (1)

(Total: 10 marks)

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